Huawei Access Node MALL-ACC-03 Optical Transport Failure SOP

Node: MALL-ACC-03 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_8687

Description

The Huawei Access Node MALL-ACC-03 is a critical component of the Malleswaram Node's optical transport network. This SOP addresses a major optical transport failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on this specific role and vendor equipment. The underlying protocol or hardware mechanics responsible for this failure domain is the Optical Transport Network (OTN) protocol, which relies on the integrity of SFP/XFP modules to transmit and receive data. When a SFP/XFP degrades or a fiber cut occurs, the OTN protocol can no longer guarantee reliable data transmission, resulting in a major optical transport failure. This failure can be further exacerbated by issues with the Rx/Tx power loss, which can cause the SFP/XFP modules to malfunction or become desensitized, leading to a cascade of errors and ultimately, a major network outage. Therefore, it is essential to have a clear and well-documented SOP in place to address this type of failure and minimize downtime.

Pre-requisite Checks

- Verify that the remote hands are dispatched to the Malleswaram Node to perform any necessary physical checks or repairs.

- Check the status of the redundant link to ensure it is functioning properly and can take over in case of a failure.

- Gather all necessary information and documentation related to the failure, including alarm logs and system configurations.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display transceiver info, Expected Output: Transceiver info for Slot 1: SFP-10G-SR-1GE, Description: This command displays the transceiver information for a specific slot, including the type, speed, and status of the transceiver. This can help identify if the SFP/XFP module is malfunctioning or degraded.
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```
Command: display optical-traffic statistics, Expected Output: Optical traffic statistics for Interface 1/1/1: 12345 bytes/s, Description: This command displays the optical traffic statistics for a specific interface, including the number of bytes transmitted or received per second. This can help identify if there are any issues with the fiber or SFP/XFP module.

Command: display alarm-info, Expected Output: Alarm info for ALARM_8687: Major optical transport failure, Description: This command displays the alarm information for a specific alarm code, including the severity and description of the alarm. This can help confirm that the issue is a major optical transport failure.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace transceiver slot 1, Description: This step involves replacing the SFP/XFP module in slot 1 with a new one. This can be done using the command 'replace transceiver slot 1' and following the on-screen instructions.

Step 2: Step: Check and repair the fiber, Command: display fiber-status, Description: This step involves checking the status of the fiber and repairing any damage or issues found. This can be done using the command 'display fiber-status' and following the on-screen instructions.

Step 3: Step: Adjust the Rx/Tx power loss, Command: adjust power-adjust, Description: This step involves adjusting the Rx/Tx power loss to ensure it is within the acceptable range. This can be done using the command 'adjust power-adjust' and following the on-screen instructions.

Post-Check Verification

```
Command: display transceiver info, Expected Output: Transceiver info for Slot 1: SFP-10G-SR-1GE, Description: This command is used to verify that the SFP/XFP module has been replaced properly and is functioning correctly.

Command: display optical-traffic statistics, Expected Output: Optical traffic statistics for Interface 1/1/1: 12345 bytes/s, Description: This command is used to verify that the fiber is functioning correctly and the optical traffic statistics are within the acceptable range.

Command: display alarm-info, Expected Output: Alarm info for ALARM_8687: No alarm, Description: This command is used to verify that the alarm has been cleared and there are no longer any issues with the optical transport network.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore transceiver slot 1, Description: This step involves restoring the original SFP/XFP module in slot 1. This can be done using the command 'restore transceiver slot 1' and following the on-screen instructions.

- Step: Restore the original fiber settings, Command: restore fiber-status, Description: This step involves restoring the original fiber settings and configurations. This can be done using the command 'restore fiber-status' and following the on-screen instructions.

- Step: Restore the original Rx/Tx power loss settings, Command: restore power-adjust, Description: This step involves restoring the original Rx/Tx power loss settings. This can be done using the command 'restore power-adjust' and following the on-screen instructions.